



# Improved mechanical properties and surface wettability of TiO<sub>2</sub> coating by SiO<sub>2</sub> reinforcement and alkali treatment



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## Abstract

This study examines the synergistic effect of SiO<sub>2</sub> particle impregnation and alkali treatment on the mechanical and corrosion properties of TiO<sub>2</sub> coatings prepared by plasma electrolytic oxidation (PEO) on commercially pure titanium (CP-Ti). The coating morphology was examined using SEM, while its composition was analyzed by EDS, XRD, and XPS. The mechanical properties were evaluated by hardness and wear tests, and the corrosion resistance was assessed electrochemically. Dispersing up to 7at% SiO<sub>2</sub> within the coating improved hardness and wear resistance by reinforcing the TiO<sub>2</sub> matrix, reducing porosity by 44% and roughness by 36%, and promoting a higher degree of crystallinity and phase stability. Consequently, the corrosion resistance increased tenfold. Alkali treatment further enhanced the mechanical properties by sealing pores and densifying the coating, most importantly, markedly improving surface wettability. Although it selectively dissolved amorphous phases in the outer layer,

causing greater surface irregularity, the coating maintained electrochemical stability, with a corrosion current density in the order of  $10^{-9}$  A/cm<sup>2</sup> and a corrosion potential of +0.17 V<sub>Ag/AgCl</sub>. The findings demonstrate that the combination of SiO<sub>2</sub> nanoparticle reinforcement and alkali treatment is an effective approach in producing compact, mechanically robust, and hydrophilic PEO coatings on Ti surfaces.



## Keywords

SiO<sub>2</sub>; Additive; Alkali treatment; Hardness; Corrosion

## 1. Introduction

Titanium (Ti) metal has become a popular material in biomedical applications, particularly in orthopaedic and dental implants, due to its excellent biocompatibility, bioinert nature, and favourable mechanical properties [1]. However, despite these advantages, Ti suffers from inherent limitations such as poor wear resistance and low surface hardness, which can compromise its long-term performance in demanding, load-bearing environments [2]. To address these challenges, researchers have explored various surface modification techniques aimed at enhancing the durability and functionality of Ti-based implants [3], [4], [5], [6]. Among these techniques, Plasma Electrolytic Oxidation (PEO), also known as Micro-Arc Oxidation, has gained significant attention as an advanced and effective method for improving the surface properties of Ti substrates [7]. PEO creates hard, porous, and highly adherent oxide layers, which not only strengthen the material but also improve its resistance to wear and corrosion [8]. The unique porous structure of PEO coatings presents a substantial advantage in biomedical applications, as it encourages bone tissue in-growth, improves biological integration, and offers a scaffold for additional functionalization, such as drug delivery or bioactive material incorporation [9]. However, from the corrosion viewpoints, pores become easy paths for solution penetration to the underlying metal. To further improve the functionality of Plasma Electrolytic Oxidation (PEO) coatings, researchers have explored a range of reinforcement materials, including metal oxide nanoparticles, ceramic particles, and

bioactive compounds [4,[10], [11], [12]]. These additives play a crucial role in enhancing the mechanical, tribological, and bioactive properties of PEO coatings, allowing them to meet the specific demands of diverse applications.

Alumina (Al<sub>2</sub>O<sub>3</sub>) nanoparticles have been utilized to increase hardness and wear resistance of PEO coatings, making them ideal for applications requiring extreme durability [11,13]. Likewise, rare earth base metal oxide, such as zirconia (ZrO<sub>2</sub>) and Yttria (Y<sub>2</sub>O<sub>3</sub>), has been incorporated into PEO coatings to enhance fracture toughness and thermal stability, particularly in environments exposed to high temperatures [12,14]. Hydroxyapatite (HA), a bioactive material resembling the mineral composition of human bone, has been demonstrated as a potential filler for reinforcing PEO coating [15]. Additionally, graphene oxide (GO) and carbon-based nanomaterials have emerged as promising candidates for reinforcing PEO coatings [10]. Despite the advantages offered by these alternative reinforcement materials, silicon dioxide (SiO<sub>2</sub>) stands out as the most effective and versatile additive for PEO coatings. The SiO<sub>2</sub> particles have been used in the PEO layer on Mg and Al [16,17]. However, there has been a lack of reports on the use of SiO<sub>2</sub> particles as reinforcement for TiO<sub>2</sub> coating. Its exceptional thermal stability, chemical inertness, and biocompatibility make SiO<sub>2</sub> an ideal candidate for enhancing coatings, particularly in biomedical applications where long-term implant performance is critical [18]. Studies have consistently demonstrated that SiO<sub>2</sub> incorporation leads to reduced porosity, improved hardness, enhanced wear resistance, and superior corrosion protection [19]. Furthermore, the SiO<sub>2</sub> ability to fine-tune the microstructure of PEO coatings ensures optimized surface properties, facilitating better biological integration and long-term reliability [17]. Given its ability to simultaneously enhance mechanical durability, biocompatibility, and corrosion resistance, SiO<sub>2</sub> remains the best choice among reinforcement materials for PEO coatings. However, the improvement of performance counterbalances the wettability. High wettability is crucial for an implant material to allow cell attachment and proliferation on its surface [20].

Alkali treatments have garnered attention for their ability to enhance surface bioactivity and corrosion resistance of metal-base implant materials. Alkali treatment, involving immersion in sodium hydroxide (NaOH), leads to surface hydrolysis and the formation of hydrated layers on the metal surface [21]. The formation of a hydrated layer helps block pathways for ionic penetration, thus protecting the implant in physiological environments [22]. The layers significantly improve the coating's biological response and physicochemical stability. Alkali-treated surfaces promote apatite nucleation and growth when immersed in simulated body fluid (SBF), facilitating faster and stronger osseointegration [23]. Alkali treatment improves hydrophilicity, which enhances protein adsorption and cell adhesion, which are critical factors for early-stage cell-material interaction [24]. Studies have shown that alkali-treated surfaces upregulate osteoblast proliferation and differentiation, directly contributing to bone regeneration [25].

The effect of alkali treatment on TiO<sub>2</sub>/SiO<sub>2</sub> coatings prepared by PEO remains underexplored. Investigating their combined or synergistic effects can provide new insights into optimizing surface performance for biomedical use, balancing mechanical durability with biological functionality. The interplay between nanoparticle incorporation and chemical modification can significantly influence the coating's microstructure and overall performance. This study aims to investigate the synergistic effects of SiO<sub>2</sub> nanoparticle incorporation and alkali treatment on the performance of PEO coatings on commercially pure titanium. The research focuses on analyzing the resultant coatings in terms of their morphology, phase composition, hardness, corrosion resistance, and surface wettability. The findings are expected to provide valuable insights for the development of advanced, multifunctional titanium coatings for biomedical and industrial applications.

## 2. Material and methods

### 2.1. Specimen preparation

A commercially pure titanium plate (CP-Ti) grade 2 with a thickness of 1 mm was used as the substrate. The plate was cut into 3 cm x 1 cm size specimens using a CNC needle-cutting machine (low-speed precision cut-off machine Minitom, Struers). A cable contact was attached at the rear surface of the specimen. The specimen was then molded in an epoxy resin, exposing one surface of 3 cm<sup>2</sup> as a working area. The specimen surface was metallographically ground using silicon carbide (SiC) paper with grit sizes #600, #1000, #1500, and #2000, followed by polishing with alumina paste until mirror finished. The specimen was degreased in acetone and then ethanol in an ultrasonic cleaner (KLS 303363), each for 3 min. The specimen was dried in a cold air stream and used immediately for the PEO.

### 2.2. Formation of PEO coating

The electrolyte used for PEO was prepared from analytical grade powder of tri-sodium phosphate dodecahydrate (Na<sub>3</sub>PO<sub>4</sub>·12H<sub>2</sub>O) and potassium hydroxide (KOH) from Merck. The composition of electrolyte was 95 g/l Na<sub>3</sub>PO<sub>4</sub>·12H<sub>2</sub>O and 2 g/l KOH. Nanopowder of SiO<sub>2</sub> (high purity, Alfa Aesar) with a concentration of 2 g/l was dispersed into the electrolyte. The ionic conductivity of the electrolyte was measured using conductivity meter (TOA Seri CM-11P). The PEO was performed in a galvanostatic mode at 300 A/m<sup>2</sup> using a DC regulated power supply (GW Instek). The electrolyte temperature was maintained at 25 °C using a regulated water bath. The process was terminated at 10 min. In the electrochemical cell, the specimen acted as an anode while a Pt mesh cylinder was used as a cathode. The voltage output during the application of a constant current to the specimen was recorded

using a bench digital multimeter (DMM, Rigol 3068). After finishing, the specimen was washed thoroughly under running tap water and then washed with DI water. The alkali treatment (AT) was performed immediately by immersing the specimen in 0.5M NaOH solution that had been heated to a temperature of 80°C. The treatment was carried out for 30min. The specimen was washed again under running tap water, followed by DI water and then dried.

### 2.3. Coating characterization

Field-Emission Scanning electron microscope (FE-SEM, Phenom Pharos) was utilized to investigate the PEO coating morphology from both surface and cross-sectional views. Energy Dispersive X-ray spectroscopy (EDS), which was attached to the FE-SEM instrument, was used to determine the composition and distribution of elements. For crystalline phases analysis, an X-ray diffractometer (XRD, Panalytical Aeris Instrument Suite) with CuK $\alpha$  source of radiation was used. The scanning speed was 0.09°/s from 2 $\theta$  of 10°–90°. The XRD data was analyzed using High Score Plus 3.0.5 software. The chemical composition of the coating surface was analyzed using an X-ray photoelectron spectroscopy (XPS, Thermo Scientific, Waltham, Massachusetts) with an AlK $\alpha$  radiation source (1486.6eV, 5mA, 15kV). The data were analyzed using CasaXPS and Peak Fit software (v 2.3.3PR 1.0). The spectra were measured and corrected after deconvolution and background subtraction (Shirley).

The coating surface roughness was investigated using a digital microscope (Keyence VHX6000). The surface roughness was represented by Ra (arithmetical mean deviation of the profile) and Rz (the point height of irregularities) values. The same microscope was also used to measure the water contact angle of a water droplet (100 $\mu$ L) on the coating surface at room temperature.

### 2.4. Mechanical properties

The micro indentation hardness was performed on the cross-section of the coating using a Vickers hardness machine to measure the coating hardness. The test procedure followed ASTM E384 [↗](#) standard using an indentation load of 25 gF and held for 10s [26]. The wear resistance was assessed using Ogoshi method with the procedure following ASTM G99 [↗](#) standard [27]. The applied load used 3.16kg of XW-41 steel ring with a diameter of 30mm, a thickness of 3mm, and an abrasion speed of 1.97m/s. The test gives a specific abrasion value, which refers to the wear rate. The specific abrasion is defined in equation (1).

$$k = \frac{V}{F \cdot s}$$

(1)

With  $k$  is specific abrasion,  $V$  is wear volume (mm<sup>3</sup>),  $F$  is normal load (N), and  $s$  is sliding distance (mm).

## 2.5. Corrosion tests

The corrosion behavior of the specimen was studied using electrochemical tests including open circuit potential (OCP), potentiodynamic polarization (PDP), and electrochemical impedance spectroscopy (EIS) using a CorrTest310 potentiostat. The tests were conducted in a 3.5wt% NaCl solution (Pro analysis from Merck). The analysis has been carried out using a three-electrode configuration with silver/silver chloride (Ag/AgCl) as a reference electrode, a platinum plate as a counter electrode, and the specimen as the working electrode. The OCP was measured for 5 min. The PDP test was performed at a potential range of 100mV below OCP up to 300mV above OCP with a scan rate of 0.5mV/s. The EIS test was conducted at a potential  $\pm 10$ mV from OCP at a frequency range of 10<sup>-1</sup> to 10<sup>6</sup>Hz. The electrochemical tests were repeated three times on three specimens treated identically.

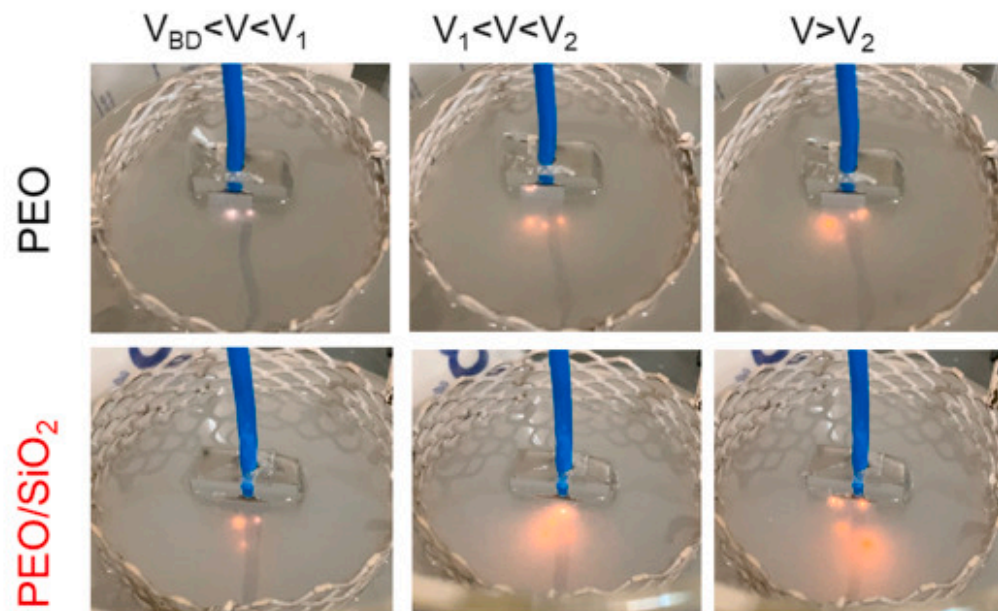
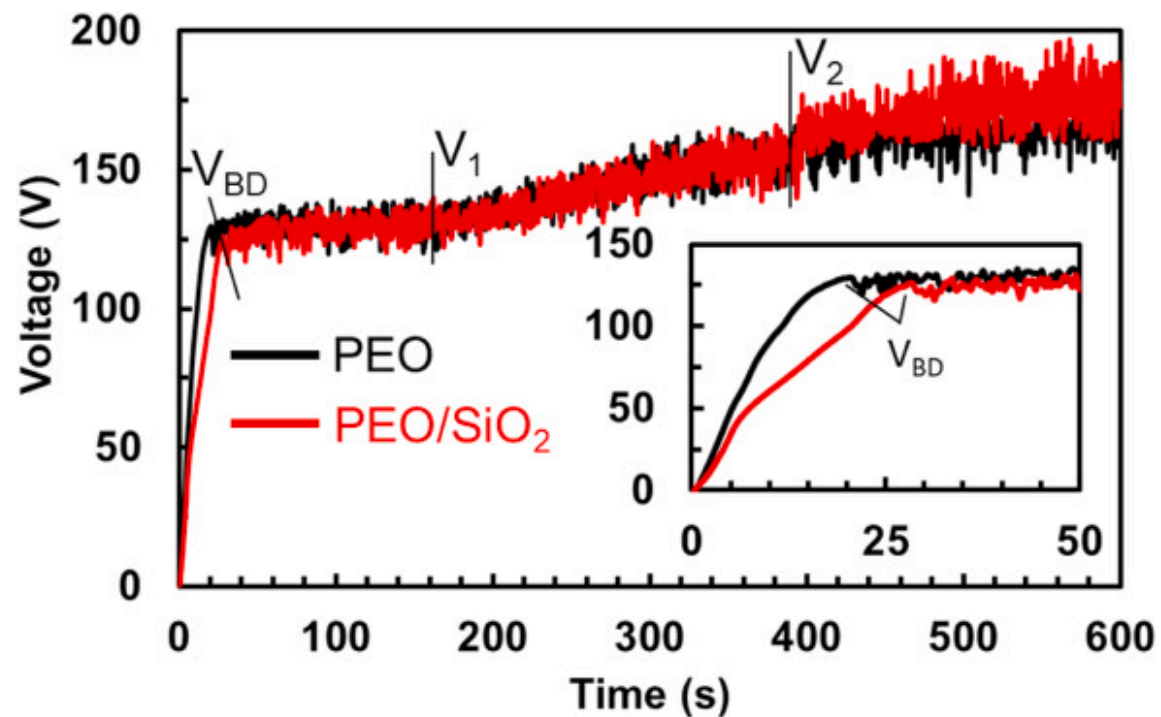
## 3. Results

### 3.1. Coating formation

The effect of dispersed SiO<sub>2</sub> particles on the voltage-time response and plasma appearance during the coating formation is shown in Fig. 1. The behavior of voltage output during PEO in the SiO<sub>2</sub>-containing electrolyte is similar to that in the original electrolyte, but there is a shift in the initial time. The curves can be divided into four stages based on the voltage gradient changes, in which the slope decreased as it progressed from stage 1–2 but then increased again towards stages 3 and 4. The slopes are listed in Table 1. The stage transition was denoted as  $V_{BD}$ ,  $V_1$ , and  $V_2$ . The breakdown voltage ( $V_{BD}$ ), which marked the transition from general anodization (stage 1) to micro-arc oxidation (stage 2), for both curves was similar at 125V. During stage 1, the Ti surface was oxidized, releasing Ti ions that further reacted with water to form oxides. A thin barrier layer was formed accordingly. The layer formation in the SiO<sub>2</sub>-containing electrolyte was slower, as indicated by the slope of 6.58V/s, than in the original electrolyte, 9.89V/s. The electric charges were then accumulated inside the layer as time progressed. Breakdown of the dielectric layer occurred at 125V, marked by the plasma sparks emission, which appeared as white sparks in Fig. 1. The  $V_{BD}$  was relatively low relative to that of reported elsewhere. The use of high conductive electrolyte effectively suppressed breakdown of

the dielectric layer. The main factor affecting the  $V_{BD}$  value is the electrolyte conductivity. Based on Ikonopisov's relation, the  $V_{BD}$  is inversely proportional to the conductivity [28]. The fact that the electrolyte conductivity,  $100 \pm 5$  mS/cm, was not altered by the addition of 2 g/l SiO<sub>2</sub> resulted in a similar  $V_{BD}$  value for PEO with and without particles. However, the presence of particles delayed the time to reach  $V_{BD}$ , which indicated an improved dielectric strength of the barrier layer relative to that of form in the original electrolyte. The particles were involved in the formation of the barrier layer. The metal oxide particles tend to collect negative charges on their surfaces and are attracted to the anode under the applied voltage by the electrophoretic force [29]. While other metal oxide nano particles such as ZrO<sub>2</sub> accelerated  $V_{BD}$  [30], similar delay time of  $V_{BD}$  was reported for the PEO in the presence of 3–10 g/l Al<sub>2</sub>O<sub>3</sub> nano particles [29]. The reason was due to a more homogeneous and denser barrier layer formed in the presence of dispersed particles.





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Fig. 1. Voltage-time response during coating formation without (PEO) and with dispersed SiO<sub>2</sub> particles (PEO/SiO<sub>2</sub>), and the plasma appearance.

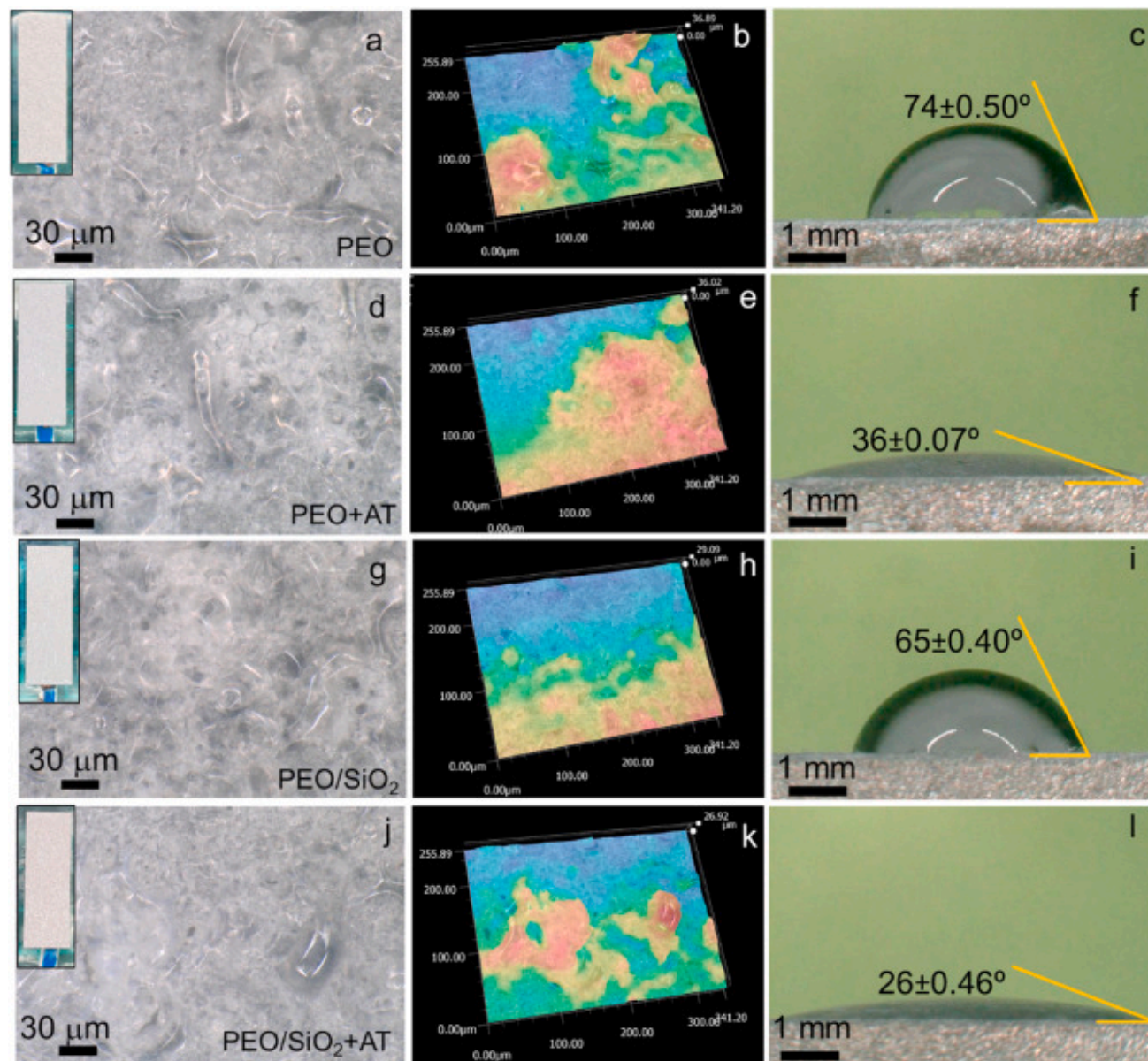
Table 1. Comparison of slope in the voltage-time response of PEO and PEO/SiO<sub>2</sub> curves.

| Stage                    | Slope for PEO (V/s) | Slope for PEO/SiO <sub>2</sub> (V/s) |
|--------------------------|---------------------|--------------------------------------|
| 1 ( $V < V_{BD}$ )       | 9.89                | 6.58                                 |
| 2 ( $V_{BD} < V < V_1$ ) | 0.04                | 0.03                                 |
| 3 ( $V_1 < V < V_2$ )    | 1.23                | 1.00                                 |
| 4 ( $V > V_2$ )          | 0.06                | 0.05                                 |

As passing the  $V_{BD}$ , the population of sparks in stage 2 increased with time. The specimen in the original electrolyte emitted whitish to yellowish plasma, while that in the SiO<sub>2</sub>-containing electrolyte was more yellowish to reddish. The color of plasma is closely related to its temperature, which means that hotter plasma emits more energetic light. The light emission was due to the thermal electromagnetic radiation and excited atomic/molecular emission. The early stage sparking discharge exhibited temperature in the range 4000–5500K, while in the later stage the temperature increases up to 5800K [31]. As the time increased, transition to stage 3 was achieved, where the plasma population decreased but the size increased. The plasma generated in the SiO<sub>2</sub>-containing electrolyte was bigger than in the original electrolyte, but the color was similar reddish. The red plasma grew further bigger, and the lifetime increased as it entered stage 4. The voltage output fluctuated severely during the last stage. The voltage spike was larger for the PEO/SiO<sub>2</sub> than the PEO curve. The disturbance in the voltage was due to complex events including formation and collapse of plasma discharges, the growth and collapse of pores, repetition of dielectric layer breakdown, disruption of electrolyte conductivity by gas (O<sub>2</sub> and H<sub>2</sub>) evolution, and local heating and cooling of the preformed oxide [32]. In all of the stages, the plasma formed in the SiO<sub>2</sub>-containing electrolyte was stronger, as observed from the size and color, than in the original electrolyte.

### 3.2. Coating morphology and composition

Fig. 2 revealed the surface profile and the contact angle test result of the coated specimens. The specimen appearance in the inset image in Fig. 2a, d, 2g, and 2j showed no significant difference between the PEO and PEO/SiO<sub>2</sub> surface before and after alkali treatment. All specimens showed a white, dull appearance. The digital microscope observation revealed a rough surface morphology on all of the specimens, as shown in Fig. 2b, e, 2h, and 2k. The growth of oxide layer involved repetition of melting and solidification of the molten oxide by the plasma discharges, producing rough morphology. The buckling of oxide gave yellow to red contrast in the image. The hills reached a maximum high (R<sub>z</sub>) of 11.36 μm for the oxide formed in the original electrolyte and lower at 8.74 μm for that formed in the SiO<sub>2</sub>-containing electrolyte. The alkali treatment enhanced the surface irregularities; the Ra and R<sub>z</sub> values were doubled for both the PEO and PEO/SiO<sub>2</sub> coatings. The surface roughness is closely related to the wettability. The contact angle measurement is aimed at determining the coating wettability. Fig. 2c, f, 2i, and 2l displayed the contact angle measurement on the coating surfaces. The measured contact angle on all sample surfaces is less than 90°, which is classified as a hydrophilic surface. The result revealed that the rougher the surface, the smaller the contact angle. The results are in line with the Wenzel model [33]. The smaller the contact angle, the higher the wettability level. The PEO/SiO<sub>2</sub> layer showed a contact angle of 65±0.40°, which was close to that of the PEO layer (74±0.50°). The post alkali treatment enhanced the coating wettability significantly, as revealed from the reduction of contact angle down to 26–36°. The enhanced hydrophilicity of the surface as a result of alkali treatment was similar to that obtained on the PEO coating formed on Mg alloy AZ31 [34]. Despite the surface roughness effect, the alkali treatment enhanced the surface polarity by the formation of hydroxide phase on the surface. Previous report revealed the formation of hydrophilic phase of sodium titanate as a result of alkali treatment of titanium oxide (TiO<sub>2</sub>) in NaOH solution [35].



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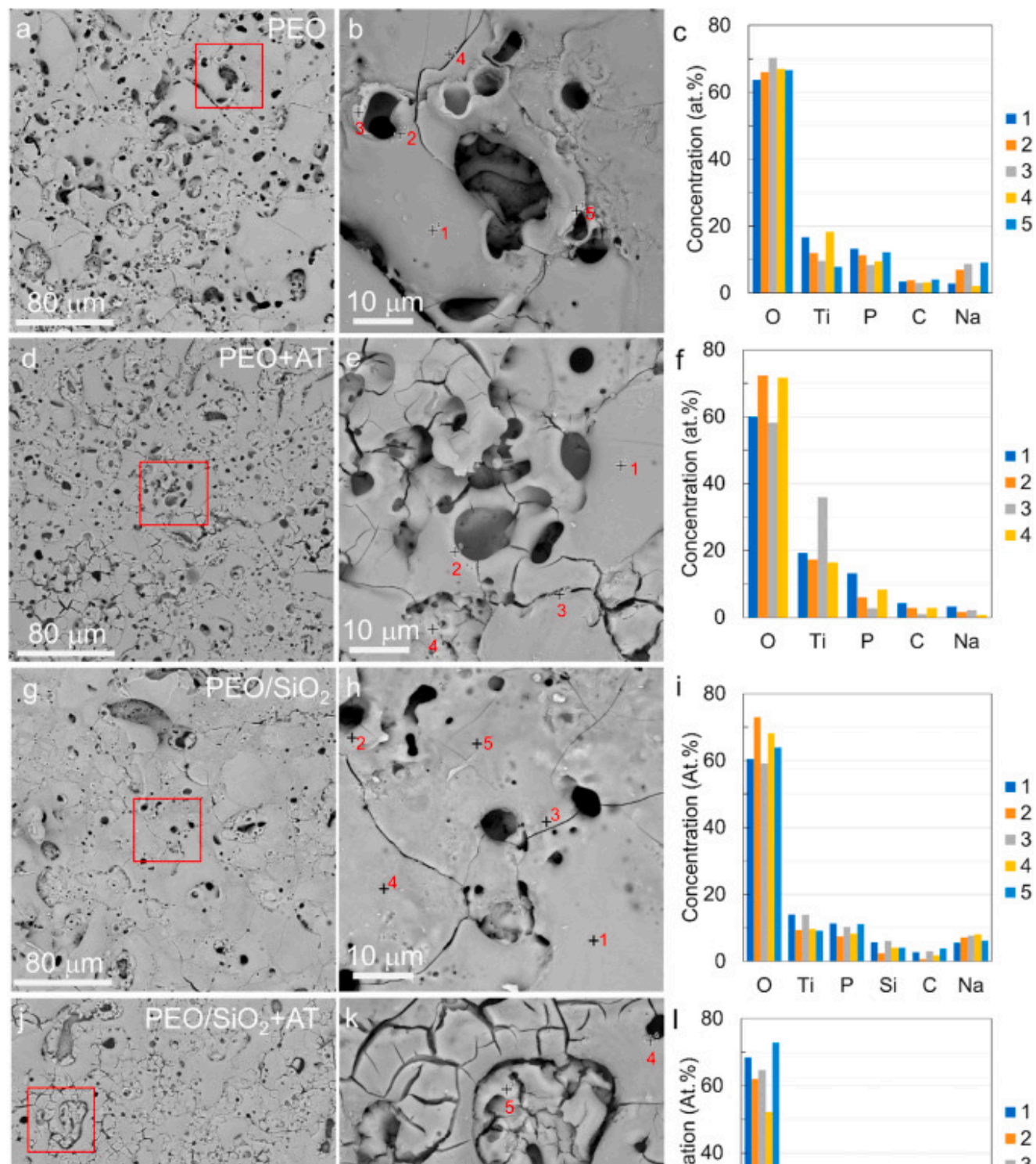
Fig. 2. (a, d, g, j) Coating morphology, (b, e, h, k) 3D profile, and (c, f, i, l) water contact angle of (a, b, c) PEO coating, (d, e, f) PEO coating after alkali treatment (AT), (g, h, i) PEO/SiO<sub>2</sub> coating, and (j, k, l) PEO/SiO<sub>2</sub> coating after alkali treatment.

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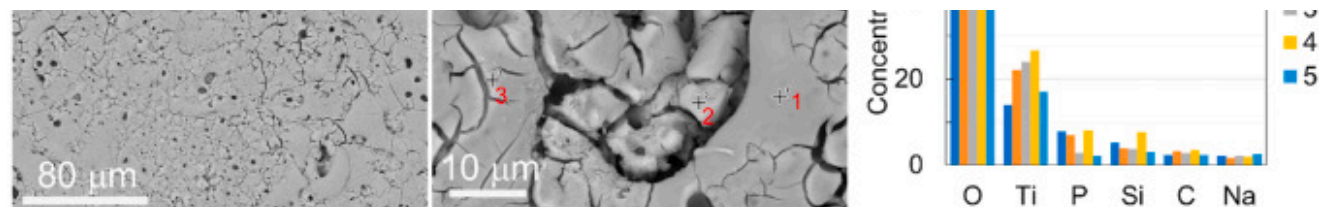
Observation of the coating surface using SEM revealed that the PEO layer exhibited a rougher surface than the PEO/SiO<sub>2</sub> surface, as shown in Fig. 3. The result is in agreement with the surface roughness investigation in Fig. 2. The PEO/SiO<sub>2</sub> coating showed significantly less pore population than the PEO coating. The SiO<sub>2</sub> particles effectively sealed the pores in the coating, similar to those reported elsewhere [19,36]. The alkali treatment resulted in a higher pore population, but the pore size was finer. The surface morphology indicated that the outermost layer had been dissolved during the alkali treatment. The cracks, in which the population increased after the alkali treatment, were formed as a result of hydration of the coating, particularly inside the SEM vacuum chamber and under the electron beam exposure. The elemental point analysis on several spots on the higher magnification images is attached to the figure. The primary elements detected in the coating were O and Ti with concentration ranges of 58–72at%, 8–35at%, respectively. The coating also contained P element with a concentration of 2–13at%. The P concentration varied from one to another site in the coating. Considering the relative ratio of O to Ti, the analysis indicated that the coating was mainly composed of titanium dioxide (TiO<sub>2</sub>). The P concentration decreased from 11 to 7at% for the PEO coating and from 10 to 5at% for the PEO/SiO<sub>2</sub> after alkali treatment. The phosphate compound was presumably reacted with the NaOH solution and further dissolved during the treatment. Other elements detected in the coatings included Na up to 9at%, which mainly concentrated around open pores. The cation Na<sup>+</sup> was incorporated in the layer by the capillary forces of the electrolyte during the coating formation, as reported earlier [37]. The Na concentration decreased after alkali treatment, possibly following the dissolution of the outermost layer. The contaminant element C was detected at a trace level of 0.7–4.2at%. The Si content in the PEO/SiO<sub>2</sub> specimen before and after alkali treatment was detected both in the bulk oxide and near the pores at 2–7at%.









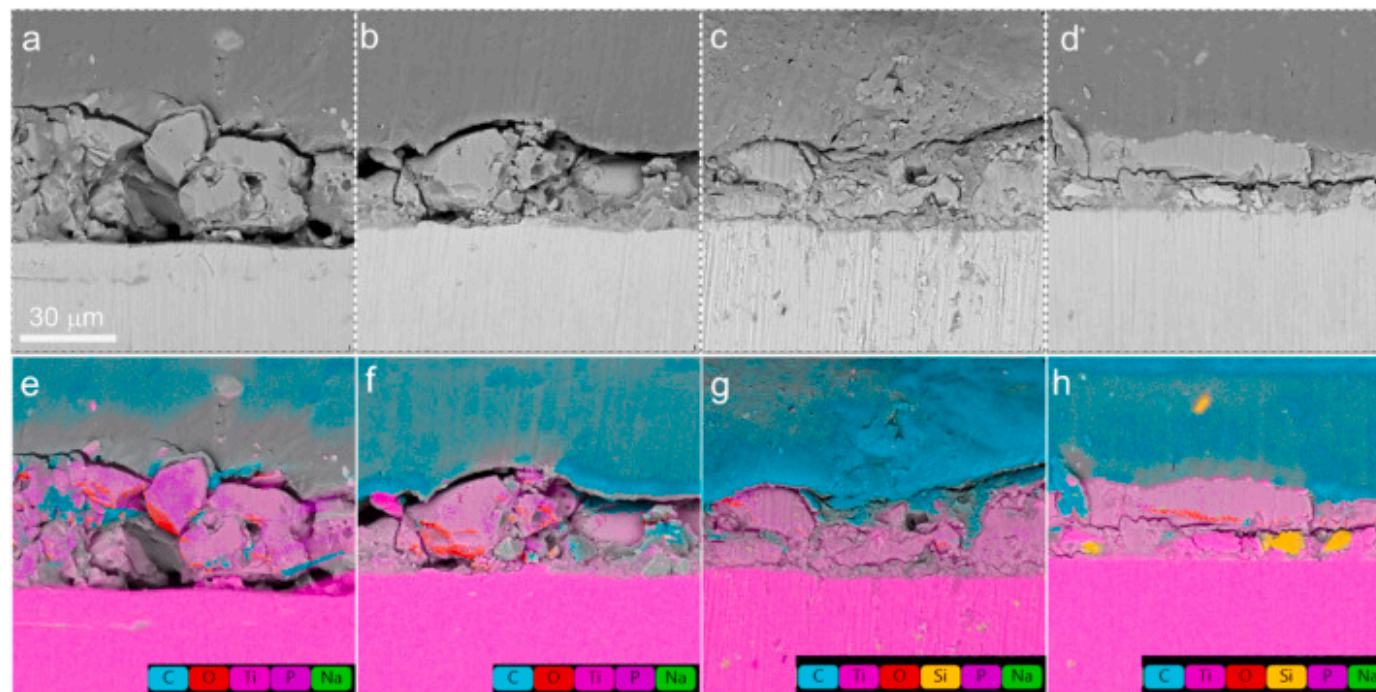


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Fig. 3. (a, d, g, j) Plane view FE-SEM images and (b, e, h, k) higher magnification images of the area inside red rectangular, and (c, f, i, l) EDS point analysis of (a, b, c) PEO coating, (d, e, f) PEO coating after alkali treatment (AT), (g, h, i) PEO/SiO<sub>2</sub> coating, and (j, k, l) PEO/SiO<sub>2</sub> coating after alkali treatment. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Fig. 4 displayed the cross-section images and the corresponding EDS elemental maps, showing the coating structure and bulk composition. The cross-section of all the coatings (Fig. 4a-d) showed cracks and voids. Wide cracks were formed during the specimen preparation, which involved grinding and polishing. The inner part of the coating, which was in contact with the substrate, was prone to crack, yielding in detachment of the coating material, leaving voids (empty space). The coating structure indicated that the inner part was more brittle than the outer part. The inner part showed strong map for a mixed Ti and O, while more purple color corresponding to P element was observed in the outer part, as shown in Fig. 4e-h. The inner part was mainly composed of Ti-O while the outer part contained Ti-O-P. Pores in a round shape were observed randomly in the bulk coating. The pores were formed as a result of gas (O<sub>2</sub> and H<sub>2</sub>) entrapment during the coating formation. The signal for Si in the map of Fig. 4g is weak, while after alkali treatment, a few areas with strong Si signal are detected in the inner part of the coating, as shown in Fig. 4h. The Si nanoparticles were smeared randomly in the coating.



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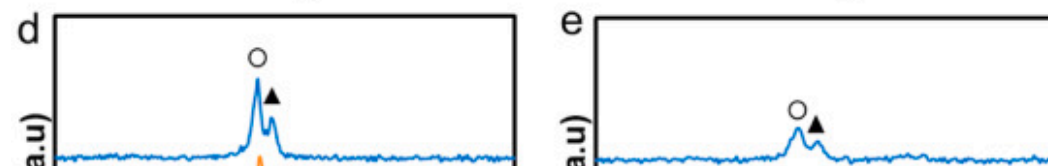
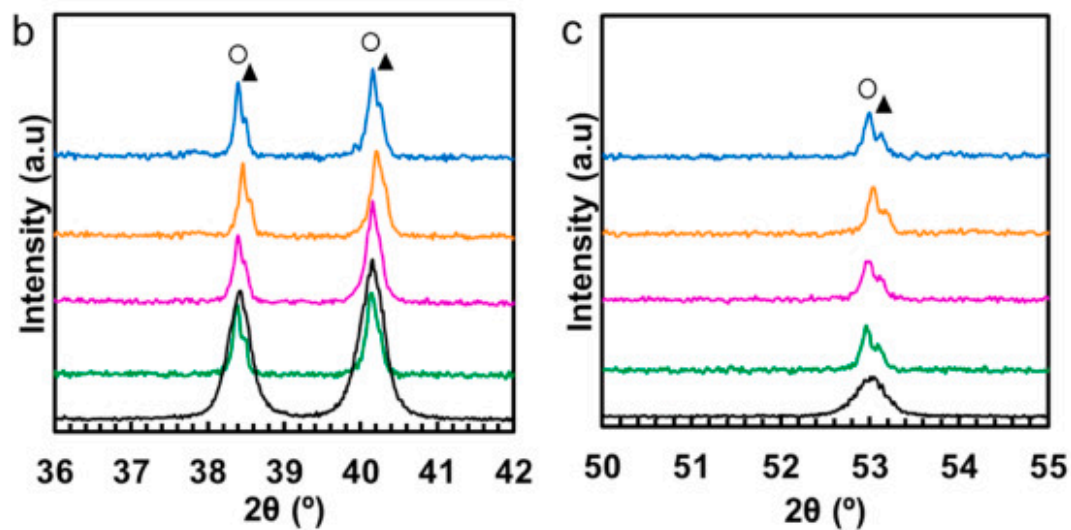
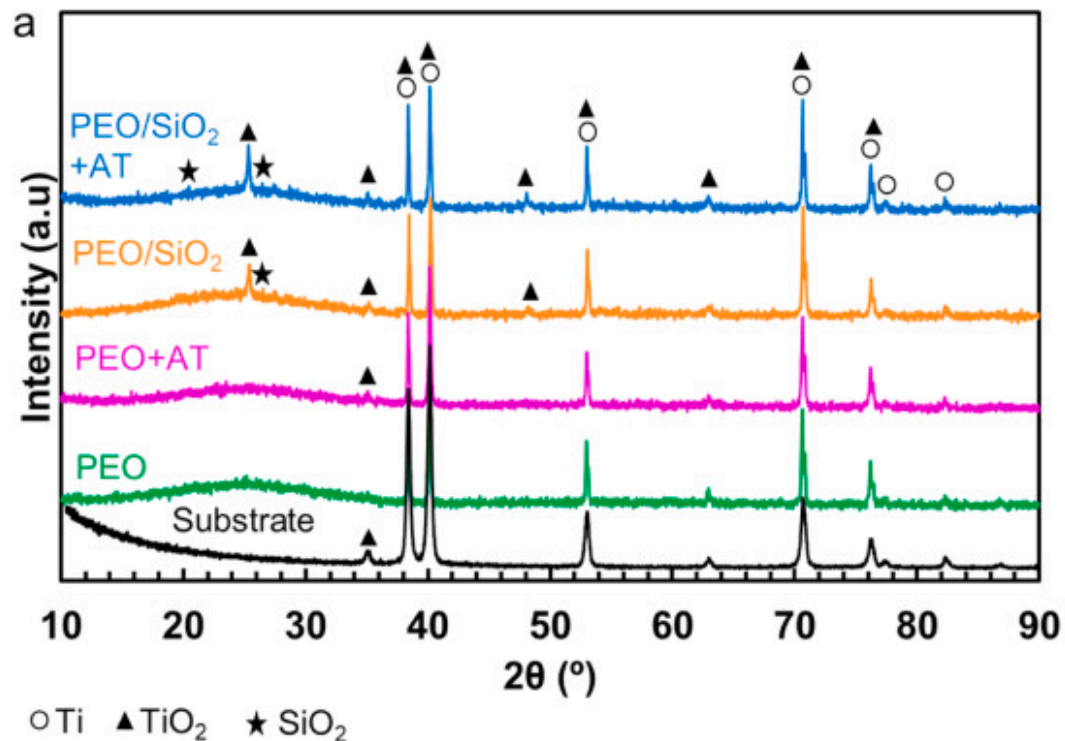
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Fig. 4. Cross-sectional view FE-SEM images and the corresponding EDS area mapping of (a, e) PEO coating, (b, f) PEO coating after alkali treatment, (c, g) PEO/SiO<sub>2</sub> coating, and (d, h) PEO/SiO<sub>2</sub> coating after alkali treatment.

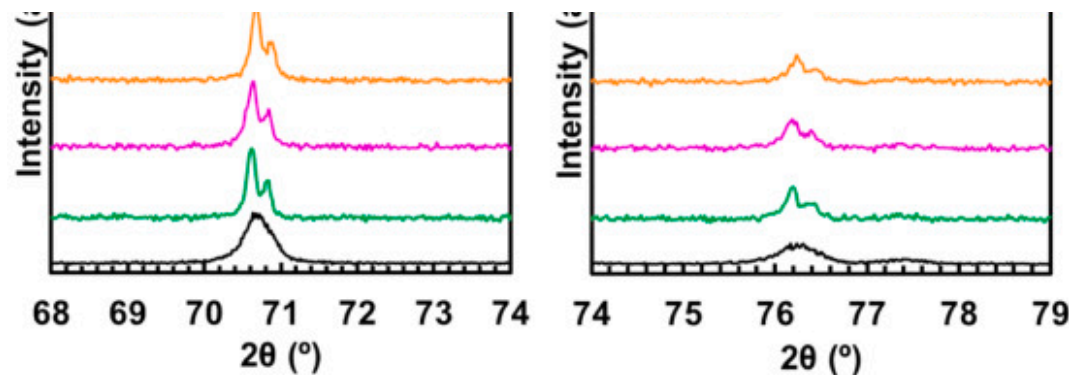
The phase composition of the coating was analyzed from the XRD pattern shown in Fig. 5. The patterns of all coatings detected a strong signal for the crystalline phase of TiO<sub>2</sub> near the Ti peaks. A small window is presented in Fig. 5b-e to clearly show the TiO<sub>2</sub> peaks. The substrate shows peaks at 38.2°, 40.1°, 53.0°, 70.4°, and 76.1°. The peaks corresponded well to Ti metal (JCPDS 44–1294). Another peak raised next to the corresponding peak in the pattern of the coated specimens was associated with the crystalline TiO<sub>2</sub> anatase phase (JCPDS 21–1272). A halo, broad diffuse peak, was observed in the coated curve in the angle between 20° and 30°, corresponding to the amorphous phase of TiO<sub>2</sub>. The unstable amorphous titanium phosphate, Ti(HPO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O, might also have contributed to the halo. The halo position was matched with the Ti(HPO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O phase (JCPDS 44–0382). The phase exists as a result of partial reduction of Ti<sup>4+</sup> to Ti<sup>3+</sup> to maintain charge neutrality due to oxygen vacancies. The peaks for SiO<sub>2</sub> phase at the angle of 20.8° and 26.6° (JCPDS 05–0490) in the PEO/SiO<sub>2</sub> and PEO/SiO<sub>2</sub>+AT patterns were quite small.

The reason is due to the low concentration and small size of the nanoparticle SiO<sub>2</sub> that is distributed randomly in the coating. A new peak raised in the halo of PEO/SiO<sub>2</sub> and PEO/SiO<sub>2</sub>+AT patterns at the angle of 25.44° corresponded to the crystalline TiO<sub>2</sub> anatase phase. The additive SiO<sub>2</sub> particles enhanced the formation of crystalline TiO<sub>2</sub> phase in the coating. Furthermore, the TiO<sub>2</sub> peaks for the PEO/SiO<sub>2</sub> (orange curve) and PEO/SiO<sub>2</sub>+AT (blue curve) specimens in Fig. 5b and c exhibited a slight shift toward higher angle, suggesting lattice contraction induced by the incorporation of SiO<sub>2</sub>. A similar rightward shift in diffraction peaks was also reported for Al<sub>2</sub>O<sub>3</sub>-TiO<sub>2</sub> composite coatings, where the addition of TiO<sub>2</sub> caused a comparable shift in the Al<sub>2</sub>O<sub>3</sub> peaks [38].









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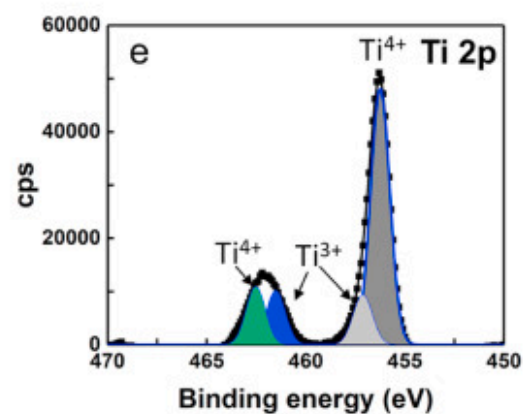
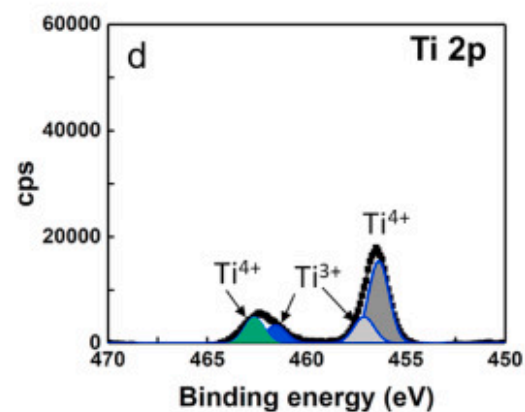
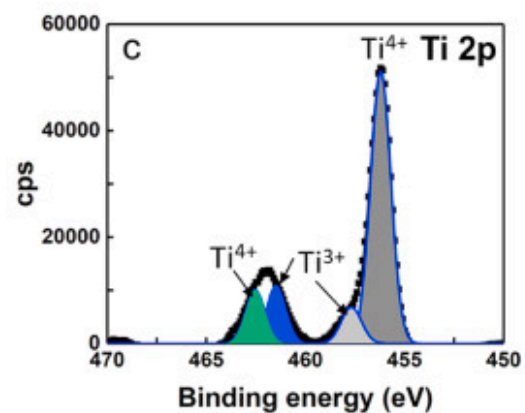
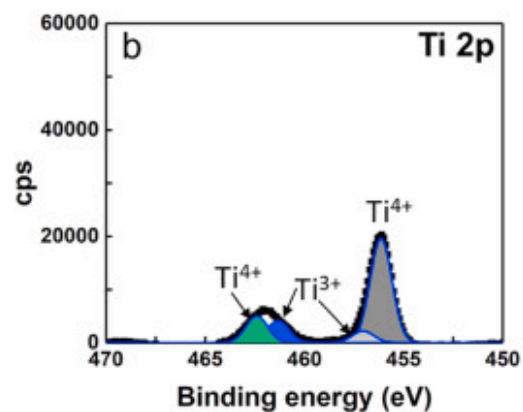
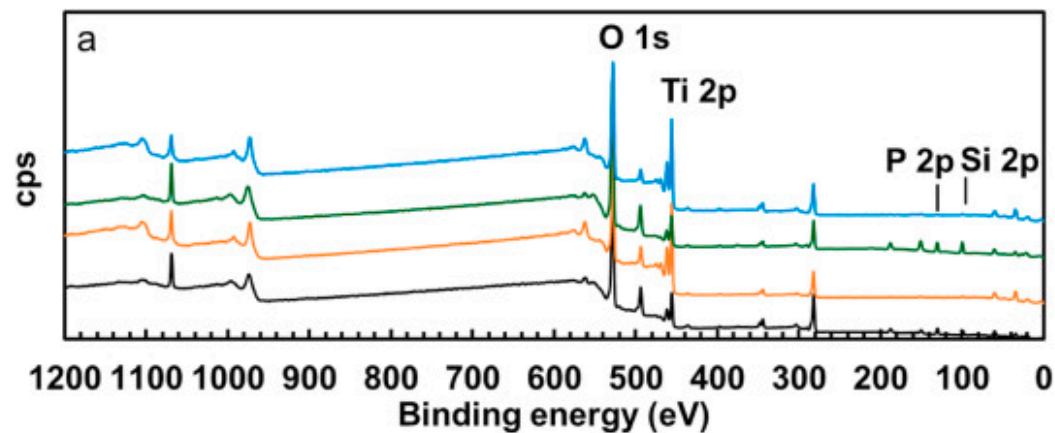
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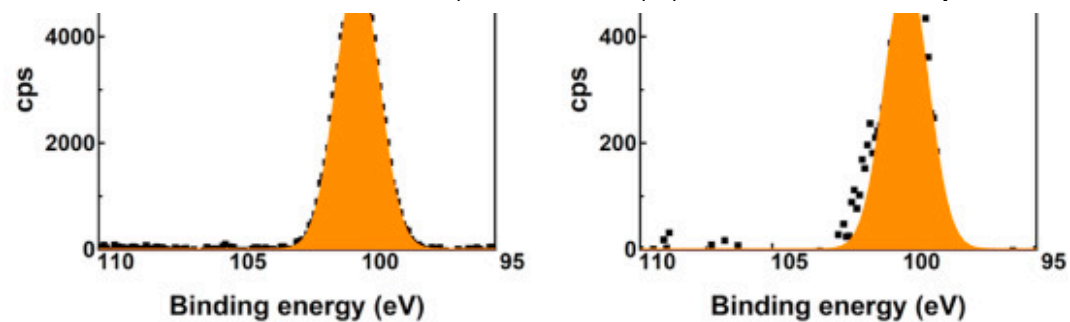
Fig. 5. a) XRD pattern showing the peaks for Ti, TiO<sub>2</sub>, and SiO<sub>2</sub> phases and b)-e) small window angle plots to zoom the TiO<sub>2</sub> peaks.

The XPS analysis was performed to confirm the phases existed in the coating. Fig. 6a displays the wide scan of XPS analysis results. The high-resolution analysis for Ti 2p is presented in Fig. 6b-e and for Si 2p is in Fig. 6f-g. The main elements detected in the coating are O, Ti, and P, confirming the EDS and XRD analyses results. Two Ti 2p peaks were observed at 456 and 462 eV, as shown in Fig. 6b-e. The band centered at 456 eV consisted of a high peak for Ti<sup>4+</sup> and a small peak for Ti<sup>3+</sup>. Another band centered at 462 eV also consisted of peaks for Ti<sup>4+</sup> and Ti<sup>3+</sup>. The P 2p peak was raised at 132 eV in Fig. 7a. The P 2p peaks were clearly resolved in the PEO and PEO/SiO<sub>2</sub> spectra, but disappeared in the spectra of PEO+AT and PEO/SiO<sub>2</sub>+AT. Moreover, the Ti 2p band was narrower, indicating the increase of Ti<sup>4+</sup> detection, after alkali treatment. The result is in line with the EDS analysis that demonstrated reduction of the P concentration after alkali treatment. The titanium phosphate phases were selectively removed

from the coatings during the alkali treatment. The Si 2p band detected in the PEO/SiO<sub>2</sub> and PEO/SiO<sub>2</sub>+AT spectra is displayed in Fig. 6f and g. The band was centered at 101.7 eV, which corresponded to Si<sup>4+</sup>, most commonly in the form of SiO<sub>2</sub> phase. The XPS analysis confirmed that the SiO<sub>2</sub> phase was embedded in the coating.





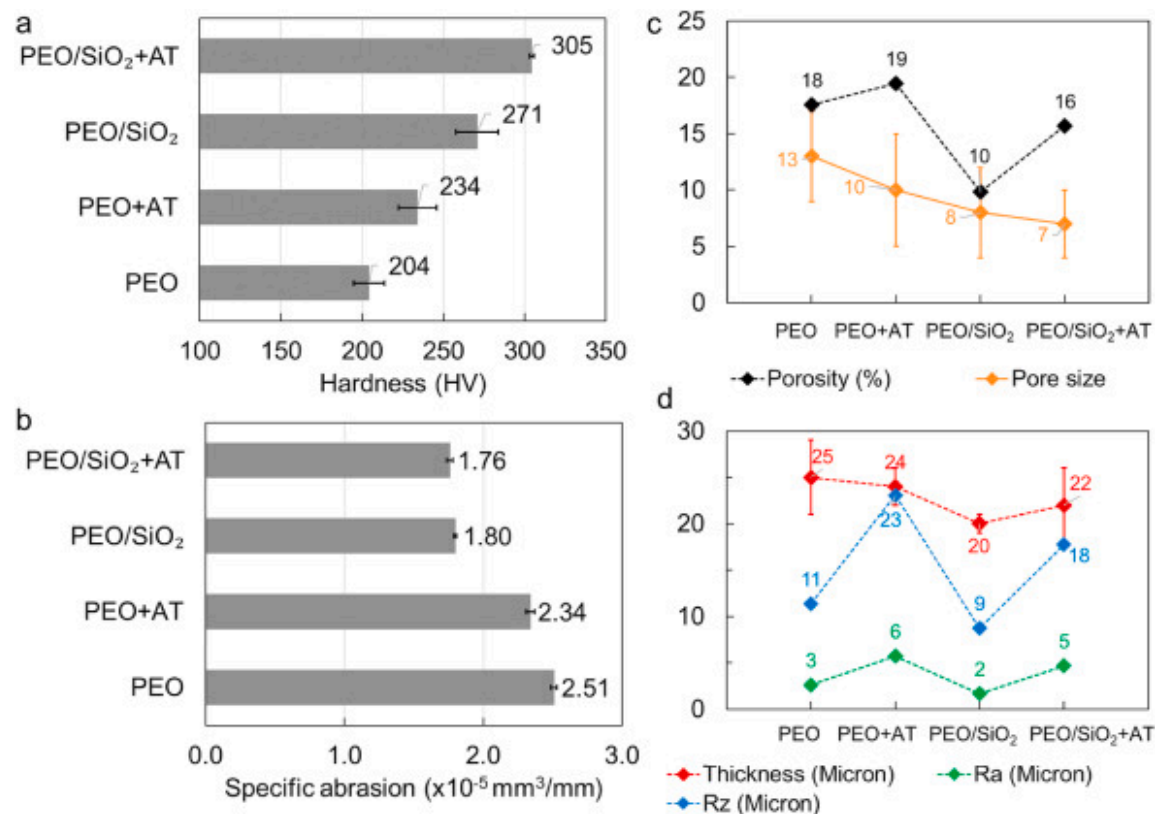


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Fig. 6. XPS spectra for a) broad survey and b)-g) window survey for Ti2p and Si2p.





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Fig. 7. Effect of SiO<sub>2</sub> particles and alkali treatment on a) hardness, b) specific abrasion, c) porosity and pore size, and d) thickness and roughness (Ra and Rz) of PEO coatings. The roughness, porosity, pore size, and thickness data are obtained from Fig. 2, Fig. 3, Fig. 4.

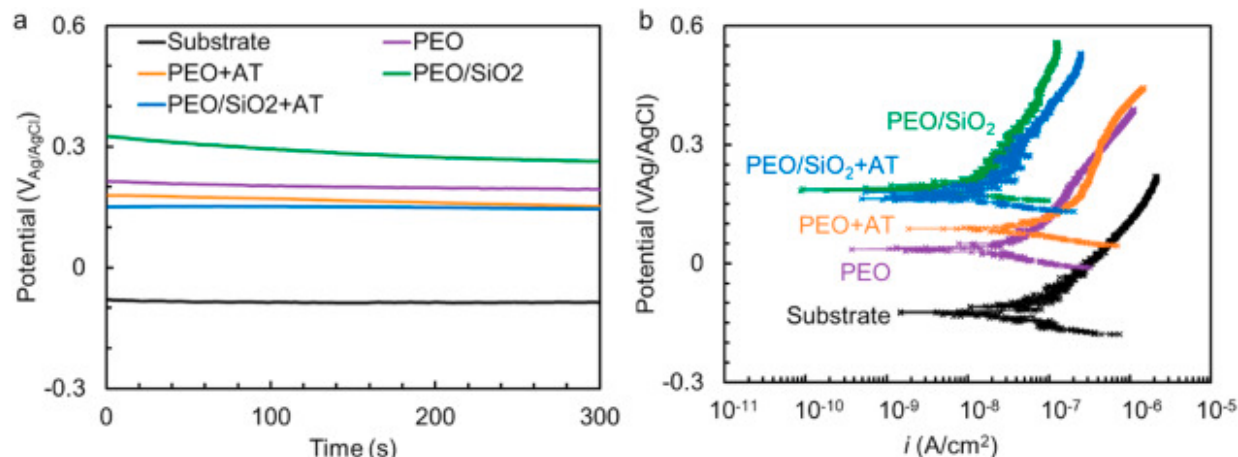
### 3.3. Mechanical properties

The mechanical properties of the coatings were analyzed from the hardness and specific abrasion measurements. Fig. 7 showed the results of hardness and specific abrasion tests. The specific abrasion data in Fig. 7b is related to the volume of material removed during the abrasive test. The specific abrasion (wear rate) is inversely proportional to the wear resistance following equation (1). Several geometrical factors affect the mechanical properties of the coating, including porosity, pore size, thickness,

and roughness, are also displayed in Fig. 7c and d. The coating hardness measured from the cross-section exhibited a value in the range of 200–300 HV. The reinforcement by SiO<sub>2</sub> nano particles increased the hardness by 67 HV. In relation to the coating geometry, the pore size and surface roughness decreased by the SiO<sub>2</sub> additive (Fig. 7c and d). Although the porosity increased, the pore size was finer after alkali treatment, enhancing the hydrophilicity. Reduction in thickness due to the SiO<sub>2</sub> additive was counterbalanced by the effect of reinforcement and pore refiner on the mechanical properties. The coating formed in the SiO<sub>2</sub>-containing electrolyte was denser and exhibited a smoother surface. Therefore, the wear rate of the PEO coating decreased from  $2.51 \times 10^{-5} \text{ mm}^3/\text{mm}$  to  $1.80 \times 10^{-5} \text{ mm}^3/\text{mm}$  as a result of SiO<sub>2</sub> addition. The individual effect of alkali treatment on the PEO coating increased the hardness by 30 HV. The synergistic effect of SiO<sub>2</sub> reinforcement and alkali treatment increased the hardness by 101 HV, exceeding the combined effect of individual contributions. The alkali treatment also improved the abrasive resistance by reducing the specific abrasion of the PEO/SiO<sub>2</sub> coating to  $1.76 \times 10^{-5} \text{ mm}^3/\text{mm}$ .

### 3.4. Corrosion properties

The effect of SiO<sub>2</sub> additive and alkali treatment on corrosion properties of the coated specimen was analyzed from the open circuit potential (OCP), potentiodynamic polarization (PDP), and electrochemical impedance spectroscopy (EIS) measurements. Fig. 8 displays the OCP and PDP curves of the coated specimen relative to the substrate. The OCP reflects the corrosion potential at an equilibrium state. The OCP indicates the susceptibility of a metal to corrode. The more active metal has more negative OCP than the noble one. The OCP in Fig. 8a was stabilized after about 100s. The substrate has a low corrosion potential at  $-0.086 \text{ V}_{\text{Ag}/\text{AgCl}}$ . The coating shifted the corrosion potential of the substrate significantly toward a positive direction, indicating improvement in the corrosion resistance. The noblest corrosion potential at  $+0.264 \text{ V}_{\text{Ag}/\text{AgCl}}$  was obtained for the PEO/SiO<sub>2</sub> specimen, followed by the PEO specimen at  $+0.194 \text{ V}_{\text{Ag}/\text{AgCl}}$ . The alkali-treated specimen exhibited a similar corrosion potential at a lower value but remained in a positive value of  $+0.146 \text{ V}_{\text{Ag}/\text{AgCl}}$ . The OCP only predicts the thermodynamic tendency of corrosion.



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Fig. 8. Effect of SiO<sub>2</sub> and alkali treatment on a) open circuit potential (OCP) and b) potentiodynamic polarization curves of PEO coatings.

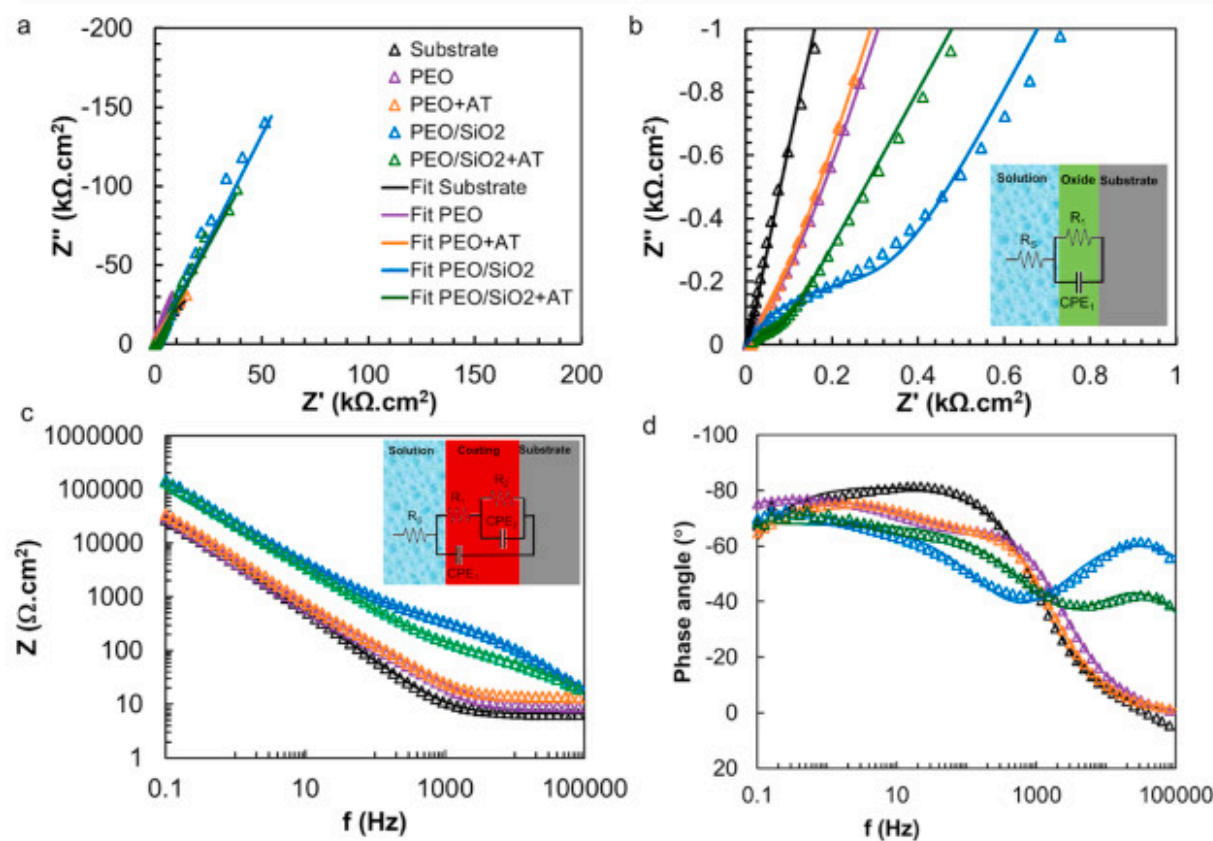
For the kinetics analysis, potentiodynamic polarization (PDP) measurement is required. Fig. 8b displayed the PDP curves of the specimens. All of the curves demonstrated a similar behavior, indicating similar corrosion mechanism. The coating does not change the corrosion behavior of the substrate but lowers the corrosion rate. The corrosion potentials ( $E_{\text{corr}}$ ) and current densities ( $i_{\text{corr}}$ ) obtained from Tafel extrapolation are listed in Table 2 together with the slope of the anodic ( $\beta_a$ ) and cathodic ( $\beta_c$ ) curves. The substrate showed a corrosion potential at  $-0.13 \text{ V}_{\text{Ag/AgCl}}$ . The corrosion potential after coating was significantly nobler in between  $+0.03$  and  $+0.19 \text{ V}_{\text{Ag/AgCl}}$ . In agreement with the OCP, the SiO<sub>2</sub>-reinforced coating showed the noblest corrosion potential of  $+0.19 \text{ V}$ , and slightly lower at  $+0.17 \text{ V}$  after alkali treatment. Most of the reports on the effect of particles (GO, Al<sub>2</sub>O<sub>3</sub>, ZrO<sub>2</sub>, CeO<sub>2</sub>) reinforced PEO revealed the corrosion potentials still in the negative values [29]. While the corrosion potential is associated with the susceptibility to corrosion, the corrosion current density is directly proportional to the corrosion rate. The current density for the substrate was  $1.11 \times 10^{-8} \text{ A/cm}^2$ . The current density value decreased by an order of magnitude in the coated specimen, indicating lower corrosion rate. The coating formed in the SiO<sub>2</sub>-containing electrolyte exhibited the lowest corrosion current density of  $1.34 \times 10^{-9} \text{ A/cm}^2$ , which was four times lower than that formed in the original electrolyte. Ti metal, which exhibits corrosion current density below  $10^{-8} \text{ A/cm}^2$ , is categorized in passivation state [39].

Table 2. Parameters obtained from potentiodynamic polarization.

| Specimen                 | $\beta_a$ (mV/dec) | $\beta_c$ (mV/dec) | $i_{\text{corr}}$ (A/cm <sup>2</sup> ) | $E_{\text{corr}}$ (V <sub>Ag/AgCl</sub> ) |
|--------------------------|--------------------|--------------------|----------------------------------------|-------------------------------------------|
| Substrate                | 50.17              | -31.21             | $1.11 \times 10^{-8}$                  | -0.13                                     |
| PEO                      | 29.63              | -16.29             | $4.49 \times 10^{-9}$                  | +0.03                                     |
| PEO+AT                   | 20.04              | -11.46             | $8.92 \times 10^{-9}$                  | +0.09                                     |
| PEO/SiO <sub>2</sub>     | 33.83              | -12.74             | $1.34 \times 10^{-9}$                  | +0.19                                     |
| PEO/SiO <sub>2</sub> +AT | 56.04              | -17.79             | $3.04 \times 10^{-9}$                  | +0.17                                     |

Fig. 9 displays the EIS results. The curve was fitted using equivalent circuit in the inset of Fig. 9b for uncoated specimen and Fig. 9c for the coated specimen. The fitted parameters are listed in Table 3. The Nyquist spectra in Fig. 9a revealed similar straight-line spectra, but the size varied. Fig. 9b enlarges the spectra at the high frequency region. The Nyquist spectra for the substrate consisted of one semicircle. The plot was modeled using a simple Randles circuit, representing the natural oxide layer. The curves for the coated specimens consisted of two semicircular arcs. The second loop is too large, so that it is seen as a nearly straight line. Larger loop diameters indicate better corrosion resistance. The equivalent circuit of the coated specimens consisted of two time-constants, following Randles circuit as reported earlier [40]. The first loop at the high to medium frequency range represents the response of the functional coating layer, while the second loop at the low frequency range corresponds to the barrier layer. The constant phase element (CPE) used in the model is associated with a deformed capacitive behavior in which the deviation from an ideal capacitor is represented by a constant  $n$ . The equivalent circuits match well with the data as revealed from the value of  $\chi^2$ , the goodness of fit, which is in the order of  $10^{-3}$ . The first loop of the PEO specimen became larger due to the SiO<sub>2</sub> reinforcement, indicating improvement in the charge transfer resistance at the electrolyte-coating interface. The coating effectively blocks the electron transfer reactions, reducing the corrosion rate. The second loop for the PEO/SiO<sub>2</sub> and PEO/SiO<sub>2</sub>+AT specimens was also significantly larger, implying substantial protection of the underneath metal substrate by the denser and more stable barrier layer than the coating formed in the original electrolyte. The alkali treatment caused a slight reduction in the corrosion resistance, as demonstrated by a mild shrinkage in the loops. In agreement, the bode impedance of the green and blue curves was the highest relative to the other specimens (Fig. 9b) in any frequency range. Furthermore, the bode phase

demonstrated a resistive-capacitive behavior, as revealed from the phase angle curves that ranged from 0° to -80°. The green and blue curves, which corresponded to the PEO/SiO<sub>2</sub> and PEO/SiO<sub>2</sub>+AT specimens, displayed more capacitive behavior as revealed from the shorter phase angle range of -40° to -75°. The  $R_2$  of the coated specimen was 2–3 times higher than  $R_1$  of the substrate, as listed in Table 3. The result indicated that the barrier layer generated during coating formation provides higher resistance than the naturally formed oxide. The values of  $R_1$ , which represented the porous coating resistance, were consistently lower than those of  $R_2$ , which represented the barrier layer resistance. The barrier layer is crucial for corrosion protection. Selective dissolution of the outermost part of the coatings during alkali treatment decreased the  $R_2$  values. However, the reduction of  $R_2$  for the PEO/SiO<sub>2</sub> (25.7 kΩcm<sup>2</sup>) was relatively mild as compared to the PEO coating (42.1 kΩcm<sup>2</sup>), indicating the stability of the SiO<sub>2</sub>-reinforced coating.



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Fig. 9. Effect of SiO<sub>2</sub> and alkali treatment on the EIS results of a)-b) Nyquist spectra, c) bode impedance, and d) bode phase of PEO coatings.

Table 3. Parameters obtained from EIS data fit.

| Specimen                 | $R_s$ ( $\Omega\cdot\text{cm}^2$ ) | $R_1$ ( $\text{k}\Omega\cdot\text{cm}^2$ ) | $\text{CPE}_1$ ( $\text{F}\cdot\text{cm}^2$ ) | $n_1$ | $R_2$ ( $\text{k}\Omega\cdot\text{cm}^2$ ) | $\text{CPE}_2$ ( $\text{F}\cdot\text{cm}^2$ ) | $n_2$ | $\chi^2$            |
|--------------------------|------------------------------------|--------------------------------------------|-----------------------------------------------|-------|--------------------------------------------|-----------------------------------------------|-------|---------------------|
| Substrate                | 2.92                               | 39.99                                      | $1.01\times 10^{-4}$                          | 0.91  | –                                          | –                                             | –     | –                   |
| PEO                      | 4.05                               | 60.89                                      | $9.63\times 10^{-5}$                          | 0.86  | 102.8                                      | $1.09\times 10^{-6}$                          | 0.99  | $5.8\times 10^{-3}$ |
| PEO+AT                   | 5.75                               | 50.89                                      | $8.92\times 10^{-5}$                          | 0.82  | 60.67                                      | $9.93\times 10^{-6}$                          | 0.99  | $4.5\times 10^{-3}$ |
| PEO/SiO <sub>2</sub>     | 2.75                               | 102.6                                      | $1.61\times 10^{-5}$                          | 0.88  | 126.9                                      | $1.05\times 10^{-6}$                          | 0.96  | $3.0\times 10^{-3}$ |
| PEO/SiO <sub>2</sub> +AT | 4.05                               | 87.45                                      | $4.64\times 10^{-5}$                          | 0.75  | 101.2                                      | $1.39\times 10^{-6}$                          | 0.76  | $1.9\times 10^{-3}$ |

## 4. Discussion

The individual and combined effects of the nanoparticles SiO<sub>2</sub> embedment and alkali treatment on the mechanical properties and corrosion resistance of CP-Ti metal have been investigated. The individual effect of dispersed SiO<sub>2</sub> particles in the PEO bath significantly enhanced the resulting coating hardness by 67 HV and decreased the wear rate nearly two times. The SiO<sub>2</sub> nanoparticles were randomly distributed in the coating with a concentration up to 7 at%, as detected by EDS. The XPS confirmed the presence of SiO<sub>2</sub> in the coating. Despite the improvement of the mechanical properties, the additive SiO<sub>2</sub> particles substantially enhanced the corrosion resistance of the coating, as revealed by the ennoblement of corrosion potentials (by 350 mV for the OCP and 160 mV for the  $E_{\text{corr}}$ ) and ten times lower corrosion current density in the order of  $10^{-9}$  A/cm<sup>2</sup>, which was considered as passive current state. In agreement, the EIS data revealed an order magnitude higher impedance for the coating formed in the SiO<sub>2</sub>-containing electrolyte. The improvement on the coating performance was related to the changes in the coating composition and structure. The PEO coating is composed of crystalline TiO<sub>2</sub> anatase and amorphous TiO<sub>2</sub> with a small fraction of Ti(HPO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O phase, as detected by XRD and confirmed by XPS analysis. The TiO<sub>2</sub> was formed as a result of reaction



Ti<sup>4+</sup> with OH<sup>-</sup> ions following reaction (2), while TiH(PO<sub>4</sub>)<sub>2</sub> phase was formed following reactions (3)–(6).



The dense and tightly packed structure of crystalline titanium oxide (TiO<sub>2</sub>) has higher hardness than that of titanium phosphate (TiH(PO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O), which has a layered phosphate structure with interlayer hydrogen bonding [41]. The TiH(PO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O phase decomposes or dehydrates at lower temperature, particularly when it exists in hydrated or amorphous state. The SiO<sub>2</sub>-containing electrolyte produced a higher concentration of crystalline TiO<sub>2</sub> phase because of the stronger plasma discharge generated than that in the free-particle electrolyte. Furthermore, a denser and more homogenous barrier layer with a higher dielectric strength was formed in the SiO<sub>2</sub>-containing electrolyte, delaying the time to reach the breakdown voltage of about 8s. The slower growth rate of the barrier layer formed in the SiO<sub>2</sub>-containing electrolyte resulted in a more uniform and compact layer. The higher concentration of crystalline TiO<sub>2</sub> phase created a more stable coating during the mechanical and corrosion tests.

The individual effect of alkali treatment enhanced the hardness and wear resistance of the PEO coating. During alkali treatment, the TiH(PO<sub>4</sub>)<sub>2</sub>·H<sub>2</sub>O phase in the outermost layer reacted with NaOH solution and then dissolved into the solution, exposing the underneath TiO<sub>2</sub> crystalline phase. EDS point analysis detected reduction of P concentration up to 12at%. Porous and loosely bonded phosphate frameworks are more susceptible to alkali attack by releasing PO<sub>4</sub><sup>3-</sup> into the solution. The Ti<sup>4+</sup> ions may react with NaOH to form sodium titanate phase [41]. Sodium titanate is a bioactive phase commonly formed by treating Ti alloys in NaOH solution, where the naturally formed TiO<sub>2</sub> layer on the surface reacts with the solution through diffusion, resulting in the formation of sodium titanate [42]. The product was in the form of a loose, porous, unstable hydrogel layer. Previous study [43] has shown that the formation of sodium titanate phase on the surface decreases the corrosion potentials of CP-Ti and Ti-Al-V alloys. Similarly, after alkali treatment, the PEO and PEO/SiO<sub>2</sub> specimens exhibited lowered OCP value at +0.146 V<sub>Ag/AgCl</sub>. However, most of the product reaction was apparently dissolved during rinsing because EDS detected reduction in the Na concentration on the surface of PEO coatings after alkali treatment (Fig. 3). The remaining sodium titanate product on the outermost surface was

easily hydrated, showing a cracked structure. The alkali treatment did not alter the PEO coating thickness significantly, but it created surface irregularities and increased the surface porosity. Part of the fine discharge tunnel, which was previously buried by the amorphous oxide, was exposed. Yet, the alkali treatment allowed the alkali species to deposit inside the pore, yielding a more compact coating. Therefore, the hardness increased by 34 HV and the specific abrasion decreased by  $1.7 \times 10^{-6} \text{ mm}^3/\text{mm}$ . The higher roughness increased the surface wettability, as demonstrated by the drastic reduction in the water contact angle value from  $74^\circ$  to  $36^\circ$ . The high surface wettability is beneficial for strong cell attachment on implant material [2]. From the corrosion viewpoint, the corrosion rate is greatly affected by the surface geometry [44]. The higher surface roughness as a result of alkali treatment provides higher surface area for the oxidation reaction. Therefore, the alkali treatment caused an increase in the passive current density of the PEO coating. However, the reduction in the corrosion resistance due to the alkali treatment is considered insignificant. The corrosion current density remained in the passive region in the order of  $10^{-9} \text{ A/cm}^2$ , and the corrosion potential remained in the positive values of  $+0.09 \text{ V}_{\text{Ag/AgCl}}$ . Moreover, the measured Bode impedance of the PEO-coated specimen after alkali treatment overlapped with that of before the treatment in all frequency ranges, suggesting similar corrosion resistance. The alteration in the passive current density was mainly due to the surface geometry factor.

The combination of SiO<sub>2</sub> reinforcement and alkali treatment had a synergistic effect on the mechanical properties of the PEO coating. Both the hardness and wear resistance increased higher than that gained by the individual effect. The PEO/SiO<sub>2</sub> coating structure was denser and harder due to the increased TiO<sub>2</sub> crystalline phase content. The alkali treatment filled the pores, further enhancing the coating compactness. The hardness increased by 101 HV and the wear rate decreased by  $7.5 \times 10^{-6} \text{ mm}^3/\text{mm}$  relative to the PEO coating. The effect of alkali treatment on the PEO/SiO<sub>2</sub> coating was similar to that on the PEO coating, but milder. The PEO/SiO<sub>2</sub> coating contained higher crystalline TiO<sub>2</sub> phase, which was more stable in the alkali solution. The reaction between the coating and alkali solution was supposed to be less than that of the PEO coating. Therefore, the effect of alkali treatment on the corrosion resistance was insignificant. The corrosion potential and corrosion current density of the PEO/SiO<sub>2</sub> were altered after alkali treatment, but the changes were small. The corrosion potential was 20 mV lower and the corrosion current density was  $1.7 \times 10^{-9} \text{ A/cm}^2$  higher than that of before treatment. The Bode impedance curves of the PEO/SiO<sub>2</sub> coating were also not affected significantly by the alkali treatment. Nevertheless, the surface wettability of the PEO/SiO<sub>2</sub> specimen increased significantly after alkali treatment, reducing the water contact angle down to  $26^\circ$ . The synergistic effect of SiO<sub>2</sub> particles and alkali treatment proved to enhance the mechanical properties and surface wettability without degrading the corrosion resistance of the PEO coating on Ti significantly.

## 5. Conclusion

The PEO coating formed on the CP-Ti substrate in a phosphate-based solution primarily consisted of TiO<sub>2</sub>. The effect of SiO<sub>2</sub> nanoparticles incorporation and alkali treatment can be summarized as follows:

- SiO<sub>2</sub> nanoparticles promoted the formation of a higher crystalline TiO<sub>2</sub> phase and produced a more compact coating with a more uniform functional and barrier layers, resulting in a drastic improvement of the mechanical hardness and wear resistance.
- Alkali treatment further increased coating compactness by filling surface pores, enhancing hardness, wear resistance, and surface wettability, while maintaining the passive behavior of Ti specimen.
- The combination of SiO<sub>2</sub> addition and alkali treatment produced a synergistic effect, delivering superior mechanical properties and wettability. Corrosion resistance was primarily improved by SiO<sub>2</sub> incorporation, with alkali treatment maintaining stability despite minor surface irregularities.

## CRedit authorship contribution statement

**Sri Rahmadani:** Methodology, Investigation, Formal analysis. **Anawati Anawati:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition, Data curation, Conceptualization. **Muhammad Dikdik Gumelar:** Investigation, Formal analysis, Data curation. **Toto Sudiro:** Software, Formal analysis, Data curation. **Ramesh Kasi:** Writing – review & editing, Validation, Resources.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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